

TA Class 1

Theory and Empirics of Economic Growth

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Macroeconomics 1 - MPE

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Introductions

Who am I?

- Frederico M P Gomes – PhD Student at Insper (4th year)
- Research on Macroeconomics
 - » Firm Dynamics under Market Frictions
 - » Credit Markets more generally
- I'm always available through e-mail: fredericompg@al.insper.edu.br in case of questions, doubts, existential crisis, etc

This Course:

- First course in macroeconomics
- Goal is to give a broad look at modern macroeconomic theory and research at a more rigorous level
- Stepping stone for anyone interested in doing serious research-adjacent work in macro

Solow Model

Solow Model

All models are wrong, but some are useful

- Box (1979)

- Solow (1956): Not worried about where growth comes from. Just whether it is possible to be sustained even amidst higher capital and labor usage over the years
- Simplest model to think about macroeconomics in a dynamic and general equilibrium framework
- Standard modeling assumptions and simple framework are enough to create a model that fits with the data (**Kaldor facts**)
- In hindsight: Most important contribution was tell us and give us a **measure of what we don't know**

The Framework: Households

Closed economy, one good, discrete time, no government, no uncertainty

What households own and earn

- State variables: capital K_t and labour L_t , both owned by households
- Both are supplied inelastically, so income is

$$Y_t = w_t L_t + r_t^K K_t$$

- Uses of income: $Y_t = C_t + S_t$

Accumulation

- $K_{t+1} = (1 - \delta)K_t + I_t$
- Closed economy with no government:
 $Y_t = C_t + I_t$, hence $S_t = I_t$

The two behavioural rules: assumed, not chosen

$$S_t = s Y_t, \quad 0 < s < 1 \text{ constant}$$

$$L_{t+1} = (1 + n) L_t$$

- No consumption–saving problem: s does not respond to r , to income, or to anything else
- No labour supply decision: L is population, and everyone works
- These are the two rules we will **microfound** later in the course

The Framework: Firms and Factor Markets

One representative firm, perfect competition, a static problem every period

Technology

- $Y_t = F(K_t, L_t)$, constant returns to scale
- $F_K, F_L > 0$ and $F_{KK}, F_{LL} < 0$
- Inada: $\lim_{K \rightarrow 0} F_K = \infty$, $\lim_{K \rightarrow \infty} F_K = 0$
- Working case: $Y_t = K_t^\alpha L_t^{1-\alpha}$, $\alpha \approx 0.35$

The firm's decision

- Rents K_t and L_t from households each period, taking w_t and r_t^K as given:

$$\max_{K_t, L_t} F(K_t, L_t) - w_t L_t - r_t^K K_t$$

Factor markets: assumed competitive

Each factor is paid its marginal product,

$$r_t^K = F_K(K_t, L_t), \quad w_t = F_L(K_t, L_t)$$

and markets clear: firm demand equals the K_t and L_t households own.

- Euler's theorem: $F_K K + F_L L = F(K, L)$, so factor payments exhaust output and profits are zero
- Under Cobb-Douglas the **shares are constant**: α to capital, $1 - \alpha$ to labour
- Net return on capital: $r_t = r_t^K - \delta$, with $r_t^K = \alpha Y_t / K_t$

Two Central Equations

Technology A_t grows at g ; per effective worker, $\tilde{y} = Y/(AL)$ and $\tilde{k} = K/(AL)$

The two central equations

$$\tilde{y}_t = f(\tilde{k}_t) = \tilde{k}_t^\alpha$$
$$\Delta \tilde{k}_{t+1} = \frac{\overbrace{sf(\tilde{k}_t)}^{\text{investment}} - \overbrace{(\delta + n + g)\tilde{k}_t}^{\text{break-even}}}{1 + n + g}$$

with $Y_t = F(K_t, A_t L_t)$, $A_{t+1} = (1 + g)A_t$ and $L_{t+1} = (1 + n)L_t$.

Setting $g = 0$ gives back the model without technological progress, in units per worker.

Steady state: $sf(\tilde{k}^{ss}) = (\delta + n + g)\tilde{k}^{ss}$

$$\tilde{k}^{ss} = \left(\frac{s}{\delta + n + g} \right)^{\frac{1}{1-\alpha}}, \quad \tilde{y}^{ss} = \left(\frac{s}{\delta + n + g} \right)^{\frac{\alpha}{1-\alpha}}$$

On the balanced growth path

$$\frac{y_{t+1}}{y_t} = 1 + g, \quad \frac{K_t}{Y_t} = \frac{s}{\delta + n + g}$$

$$r_t = \alpha \frac{Y_t}{K_t} - \delta = \alpha \frac{\delta + n + g}{s} - \delta$$

Golden rule: $f'(\tilde{k}^{gr}) = \delta + n + g$

What the Model Does Not Do

Every shortcoming is an agenda item for the rest of the course

The rules are assumed, not chosen

- s is a constant: nobody ever trades consumption today for consumption tomorrow
- The golden rule says k^{ss} *can* be inefficient, not what households would pick
- Same for labour: L grows mechanically

Growth is assumed, not explained

- Without g there is no long-run growth in y ; with g , all of it is g
- Ideas are non-rival $\Rightarrow$ increasing returns, which sit badly with perfect competition (Romer, 1990)

The cross-country predictions are too strong

- Same $F(\cdot)$ everywhere $\Rightarrow$ unconditional convergence, which we do not see
- Diminishing returns $\Rightarrow$ huge return gaps, $r_{US}^K / r_{MEX}^K \approx 9.4$. Capital should flood in, and it does not (Lucas, 1990)

Where this goes next

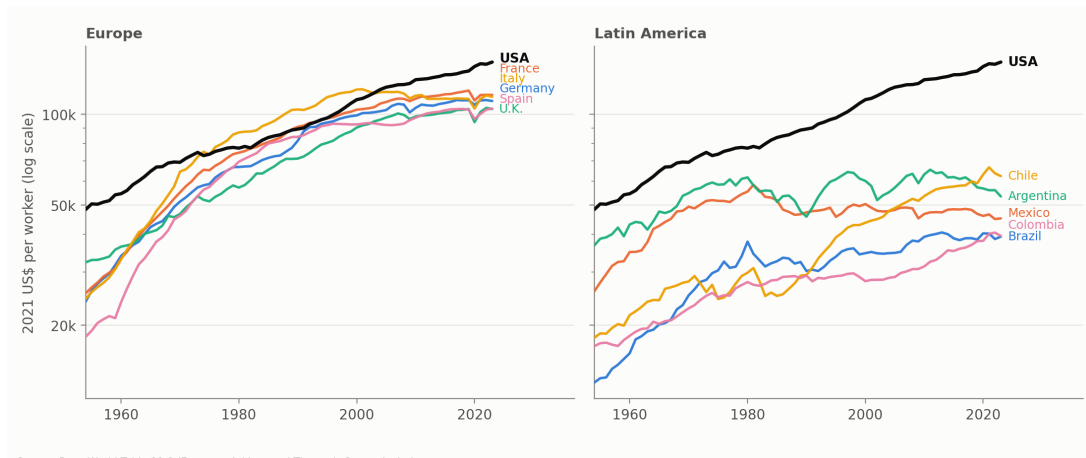
Today: take the model to the data, then measure A .

Next classes: microfound the rules, consumption and saving first.

Problem Set 1

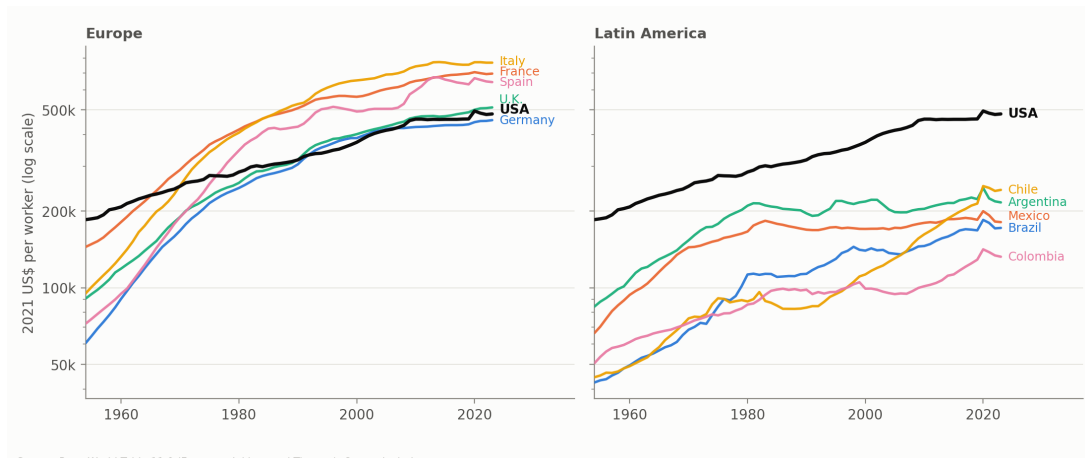
Kaldor Facts

Fact 1: Output per Worker Grows at a Roughly Constant Rate



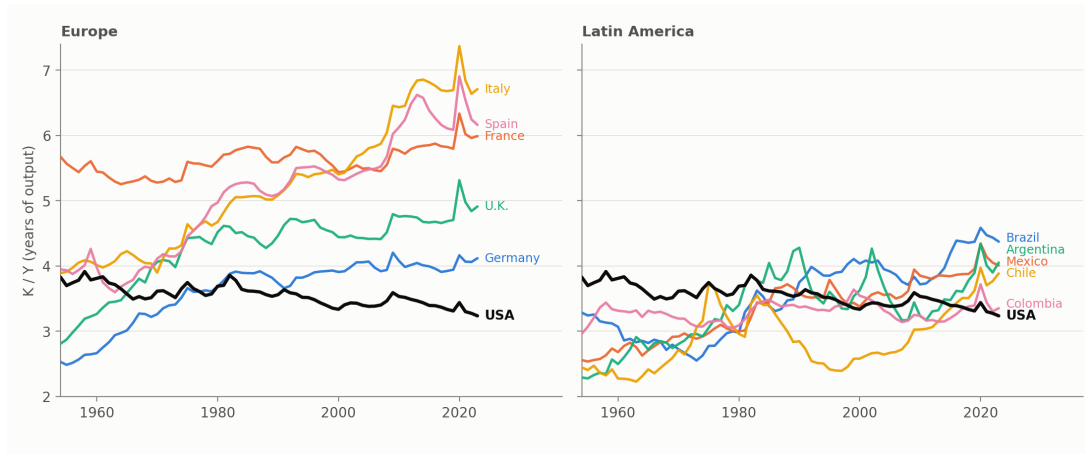
- A straight line in logs means a constant growth rate: the U.S. grew at **1.63% per year** for 70 years
- Europe is a catch-up story and Latin America is not a constant-growth story at all

Fact 2: Capital per Worker Grows at a Similar Rate

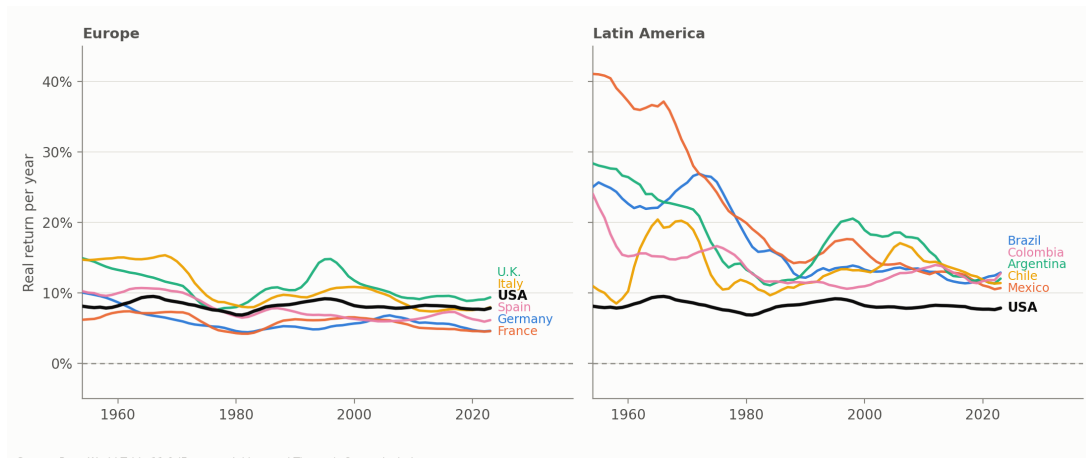


- U.S. capital per worker grows at **1.40%** against **1.63%** for output per worker: close enough that K/Y barely moves
- Europe deepened capital at 4–6% per year until 1980, and at under 1.2% since 2000

Fact 3: The Capital-Output Ratio is Trendless

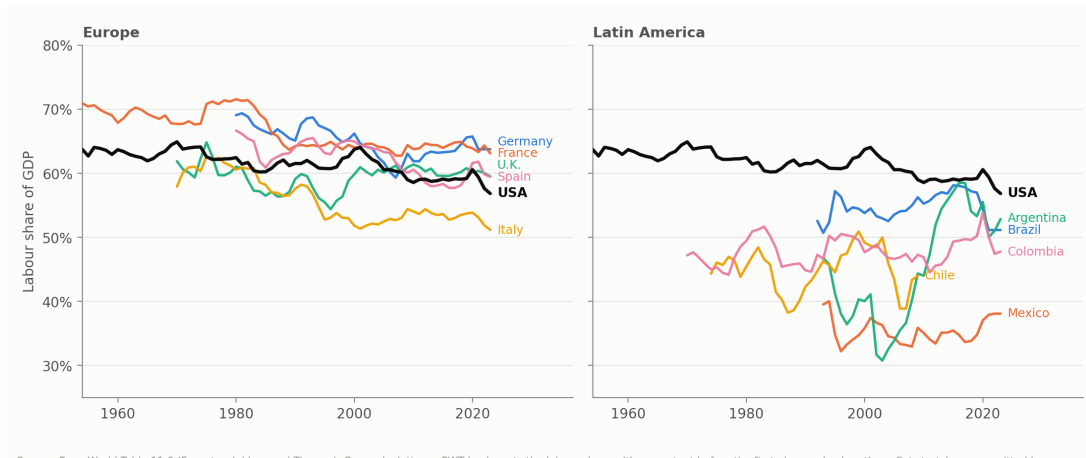


Fact 4: The Real Return on Capital Has No Trend



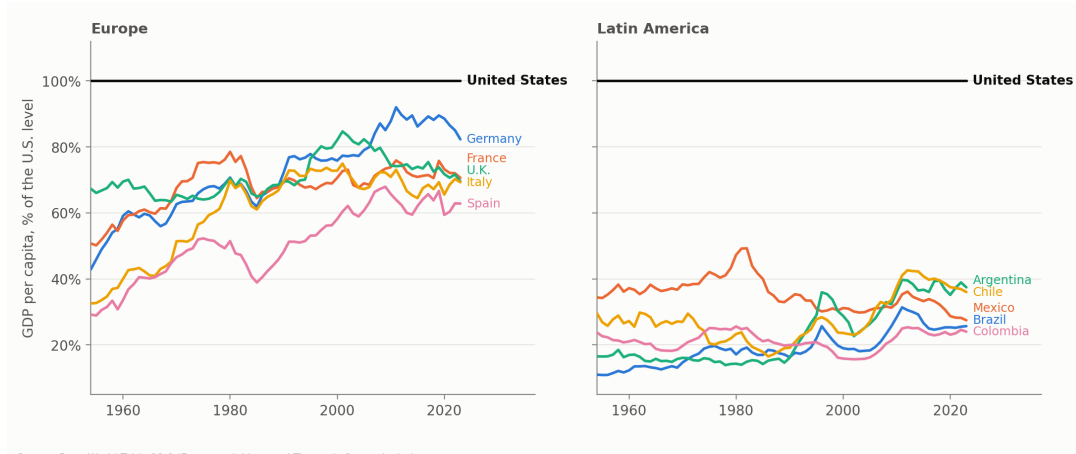
- U.S. return on capital: **8.3%**, **8.3%** and **7.9%** across 1954–79, 1980–99 and 2000–23, despite a fivefold rise in K/L
- Latin American returns start high and fall: capital deepening from a low k

Fact 5: Factor Shares Are Roughly Constant



- U.S. labour share goes from **0.63** to **0.59**, Europe sits in the same band: constant shares are what buy us Cobb-Douglas
- Latin America is systematically lower (Mexico 0.35), which is partly about measuring self-employment

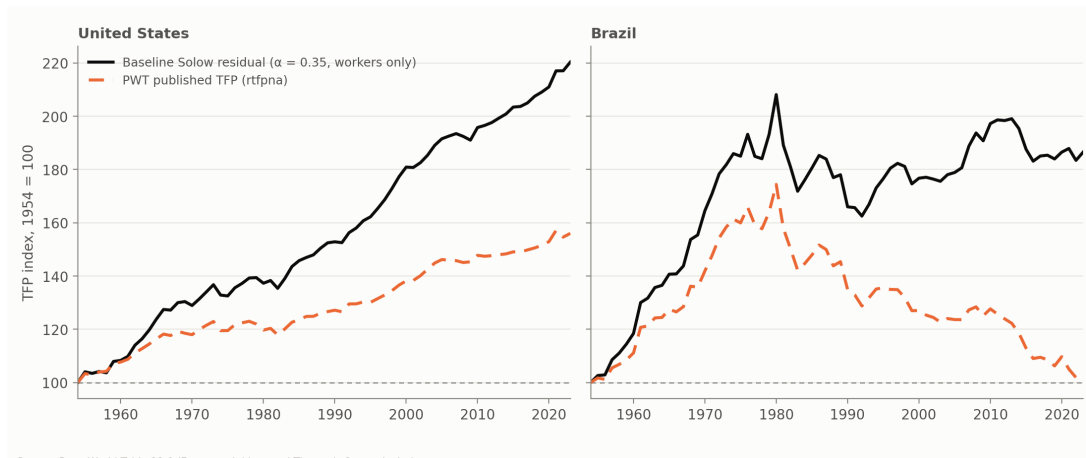
Convergence across Countries



- Europe converged towards the U.S. and then stopped, at **60–80%** of the U.S. level
- Latin America never converged: Mexico is further from the U.S. today (**28%**) than in 1960 (**37%**)

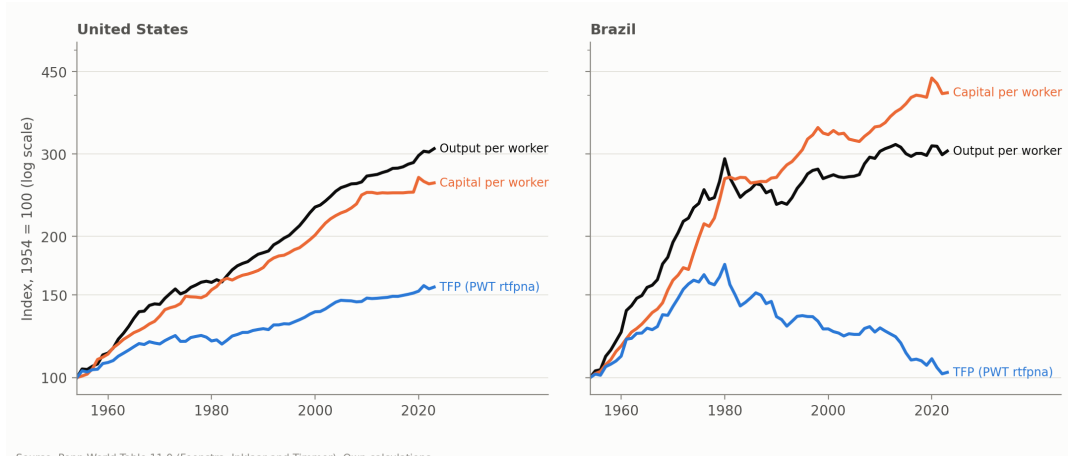
Growth Accounting

TFP Levels: the Residual Depends on What You Measure



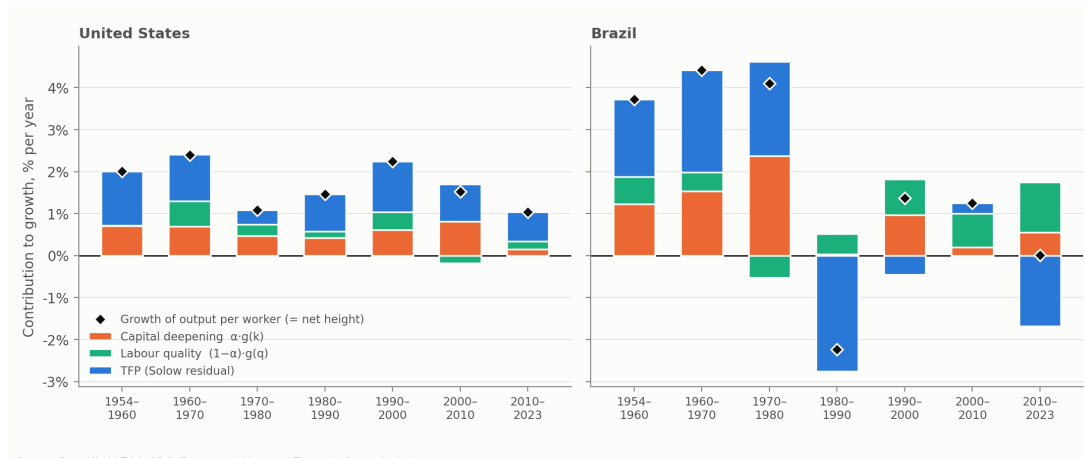
- Same data, two defensible choices: over 70 years U.S. TFP rises **120%** on the textbook residual and **56%** on PWT's published series
- For Brazil the gap is the whole story: **+87%** against **+3%**

Sources of Growth: Output, Capital and TFP per Worker



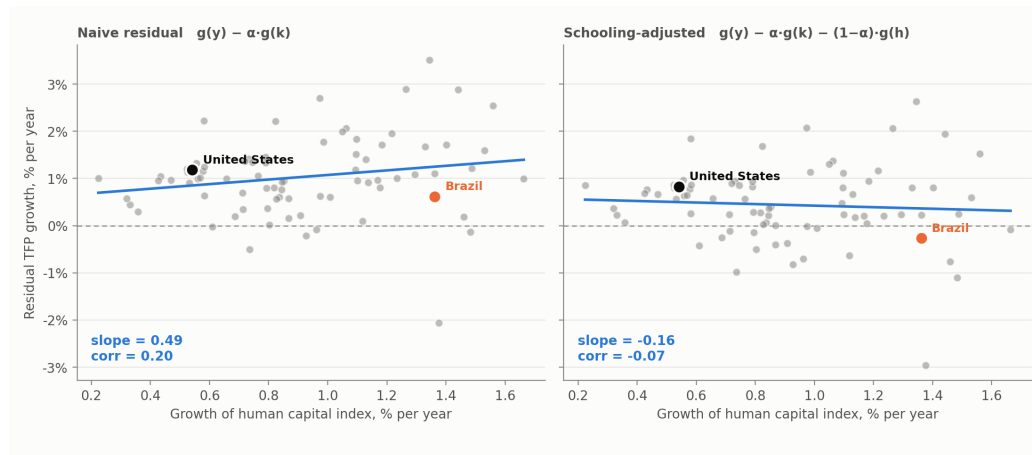
- Brazil accumulated capital *faster* than the U.S. (2.05% against 1.40% per year) and still stalled after 1980
- What stalled is the residual, not the capital stock: you cannot grow forever by accumulating

Growth Accounting: Contributions by Decade



- In the U.S. the residual is just over half of growth in every subperiod, and **63%** of it since 2000
- Brazil's 1980s subtract about **2.7 pp per year** of pure residual, with capital and schooling still contributing positively

Education and TFP



- Slope of **0.49** across 78 countries, and **0.44** in a decade panel with country fixed effects
- The two slopes differ by exactly $1 - \alpha = 0.65$: that is an identity, not a finding. And $R^2 = 0.04$, so schooling explains the residual's level, not its dispersion

Thank you!

References I